

SMART INDIA HACKATHON 2026 · PHYSICAL VALIDATION DATASET

Nirikshak Real Product Data Collection Guide

Collector Edition — a short, practical guide for gathering raw evidence from real physical packaged products.

3+ Products Minimum

10 Standard Photos

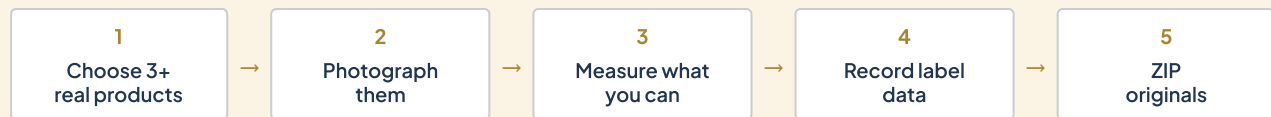
No JSON / YAML Needed

Purpose & Role

Your job is simple: collect raw evidence from real physical packaged products. You do not need to run Nirikshak, write JSON/YAML, calculate accuracy, or decide the final legal verdict — the project's rules engine and benchmark team handle that later.

Golden Rule: Record what you actually see or measure. Never guess a missing value — write "NOT AVAILABLE" or "UNREADABLE" instead.

✓ Quick Flow



1. Choose Your Products

Collect at least 3 physical packaged products. More is welcome. Prefer variety rather than three nearly identical products.

Categories

Food / beverage

Cosmetics / personal care

Pharmaceutical / consumer health

Electronics / electrical

Other packaged commodities

Packaging & surface variety

Pouch

Box

Bottle

Jar

Tube

Flat pack

Curved

Matte

Glossy

Reflective

Textured

Prefer products where important declarations are spread across different panels (not all on the front).

Do Not: Use downloaded product photos, screenshots, catalog images, or AI-generated images. Use the actual physical package only.

2. Photos — 10 Standard Images per Product

First inspect the entire package. The names below are a standard capture set; they do not mean every important label must be on the front — a required declaration can be on the front, back, left, right, top, bottom, or another label-bearing surface.

Filename	What to capture
front_01.jpg	Full main/front panel, straight-on
back_01.jpg	Full opposite/back panel — usually a high-priority declaration view
left_01.jpg	Actual left side/panel (do not crop from the front photo)
right_01.jpg	Actual right side/panel (do not crop from the front photo)
front_angle_01.jpg	Front from a moderate real angle (~20–30°)
back_angle_01.jpg	Back from a moderate real angle
glare_01.jpg	Useful label area with realistic glare/reflection
low_contrast_01.jpg	Useful label area under lower-contrast conditions
far_01.jpg	Same/important label from farther distance — a genuine distance test
close_01.jpg	Close, sharp view of an important label/declaration

Important: If a relevant declaration is on the top, bottom, cap, seal, cylindrical panel, or another area, take extra photos — e.g. top_01.jpg, bottom_01.jpg, side_02.jpg, label_01.jpg. Coverage matters more than the filename.

3. Capture Conditions

- Record phone/camera model and camera/lens if known.
- Record resolution/megapixels, aspect ratio, optical/digital zoom, flash, and HDR/AI enhancement settings if known.
- Record approximate camera-to-package distance for each photo or each capture group.
- Record lighting/background, room brightness, flash, tripod, camera angle, obstruction, and reflections when relevant.
- Keep the original image files. Avoid WhatsApp/compressed copies.

Do Not: Crop, resize, sharpen, denoise, beautify, artificially add glare, or otherwise enhance the originals. Do not call a phone photo a 1200-DPI scan.

4. Calibration / Reference Photo

Take one separate reference image when using the supplied physical reference object (for example, the project's ₹10 coin reference) — stored as `reference/calibration_01.jpg`.

Reference type	e.g. ₹10 coin, or the project-approved reference object
Known dimension & unit	The reference object's known size (e.g. diameter in mm)
Placement	Keep the object visible, in the same plane as the label, and reasonably sharp
What to record	Which reference object was used, and how it was placed

Note: A photograph contains pixels, not millimetres by itself. Do not digitally resize or alter the reference image — capture it so its boundary/ring is usable for calibration.

5. Physical Package Measurements

If you have a suitable measuring device, measure the physical package. Prefer two independent measurements where practical.

Package dimensions	Width, height, depth (or applicable dimensions)
Device	Digital vernier / caliper / ruler; model if known
Resolution / least count	Record the device specification
Raters	Names/initials of the people measuring
Repeated values	Record both readings rather than averaging away disagreement
Notes	Curvature, flexible packaging, deformation, inaccessible edge, etc.
Evidence	Photo of the measurement setup/value if practical

If You Cannot Measure: Write "NOT AVAILABLE — equipment/method unavailable". Never estimate.

6. Font / Declaration Height Measurements (if available)

For physical font-size ground truth, measure an actual visible character/text declaration using an appropriate calibrated method. Record enough detail for another person to understand exactly what was measured.

- Declaration measured (e.g. MRP, net quantity, unit sale price).
- Exact visible text/character and image filename.
- Two measurements when practical, device and least count/resolution.
- Calibration/reference used and rater(s).
- Method and notes, including curvature or difficult viewing conditions.

Do Not: Infer millimetres from pixels without valid calibration/reference evidence.

7. Record What Is Actually Printed

For each product, transcribe the visible package information as accurately as possible. Keep optical transcription separate from your interpretation.

Information	Record if present
MRP	Exact printed MRP and unit/currency presentation
Net quantity	Exact value + unit
Unit sale price	Exact printed unit sale price, if present
Manufacture/date	Month/year or other printed date information
Best-before/use-by	Where present/relevant
Manufacturer / packer / importer	Exact printed name
Address	Exact printed address
Consumer care	Phone/email/website as printed
Batch / lot	Exact printed identifier
Country of origin	For imported products, where printed
Other declarations	Anything relevant or unusual

If Unreadable: Write "UNREADABLE" rather than inventing the text. Do not silently correct spelling or rewrite what the package says.

8. Optional Difficult Images

You may preserve difficult captures separately in `images/excluded/`: steep angle, too close/out of focus, heavy shadow, duplicate, or other unusable images. Do not delete the original evidence.

9. Simple Folder Structure

Use one folder per product/SKU. Keep the original filenames and add extra images only when needed.

```

PRODUCT_01/
├── front_01.jpg
├── back_01.jpg
├── left_01.jpg
├── right_01.jpg
├── front_angle_01.jpg
├── back_angle_01.jpg
├── glare_01.jpg
├── low_contrast_01.jpg
├── far_01.jpg
├── close_01.jpg
├── reference/
│   └── calibration_01.jpg
├── images/excluded/ (optional)
└── COLLECTION_NOTES.txt

```

10. Copy-Paste Collection Notes

Fill this in with plain text. You do not need to create JSON or YAML.

PRODUCT / SKU

Brand:

Product name:

Category:

Packaging type:

Approx. purchase/source location:

Date collected:

CAMERA / DEVICE

Phone/camera:

Camera/lens:

Resolution / megapixels:

Aspect ratio:

Zoom:

Flash:

HDR / AI enhancement:

Format:

CAPTURE CONDITIONS

Lighting:

Background:

Room / outdoor:

Approx. distances:

Angle / obstruction / reflection notes:

PHYSICAL MEASUREMENTS

Width:

Height:

Depth:

Device + least count:

Rater 1 / Rater 2:

Second readings:

Measurement notes:

FONT / DECLARATION MEASUREMENTS (IF AVAILABLE)

Declaration:

Image filename:

Exact visible text:

Reading 1 / Reading 2:

Device / method:

Calibration/reference:

Rater(s):

Notes:

VISIBLE PACKAGE INFORMATION

MRP:

Net quantity:

Unit sale price:

Manufacture/date:

Best-before/use-by:

Manufacturer / Packer / Importer:

Address:

Consumer care:

Batch/lot:

Country of origin:

Other declarations:

EXTRA IMAGES & NOTES

List any top/bottom/label/side images and why they were needed.

Notes / problems:

11. Before You ZIP and Send

- ☐ At least 3 real physical products collected.
- ☐ All original photos are present and open correctly.
- ☐ The 10 standard images are present where applicable.
- ☐ Extra label photos were taken if important information was elsewhere.
- ☐ Reference/calibration image is included when collected.
- ☐ Camera/device and capture-condition notes are filled in.
- ☐ Physical measurements are recorded where equipment was available.
- ☐ Font measurements are recorded only when actually measured.
- ☐ MRP, net quantity, dates, manufacturer/packer/importer and other visible declarations are transcribed.
- ☐ Unreadable or unavailable values are explicitly marked — not guessed.
- ☐ No WhatsApp/compressed replacements, screenshots, edits, filters, or AI enhancement of originals.
- ☐ Product folder is clearly named and ready to ZIP.

What You Do Not Need To Do: Do not run Nirikshak before freezing your ground truth · Do not calculate OCR CER, scale error, font MAE, accuracy, or benchmark scores · Do not write benchmark code · Do not create JSON/YAML · Do not make the final legal compliance decision — record the evidence; the project rules engine handles legal evaluation.

Golden rule & Submission Directive

Golden rule: raw evidence first. No guessed values. No changing ground truth to match the system's output.

Submission: ZIP one product collection set per contributor. Minimum 3 products; more products and greater physical variability are useful.